

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B. Tech. Branch : Electronics/ Electronics and Telecommunication/

Electronics and Communication Engineering

Semester : III

Subject Code & Name: BTETC302/ BTEXC302 & Electronic Devices and Circuits

Max Marks: 60

Date: 07/02/2025

Duration: 3 Hours

Instructions to the Students:

1. Each question carries 12 marks.

2. Question No. 1 will be compulsory and include objective-type questions.

3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.

4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.

5. Use of non-programmable scientific calculators is allowed.

6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
	Which region of the transistor is highly doped?	CO1	
	a. Emitter		
	b. Base		
	c. Collector		
	d. Both Emitter & Collector		
	What is the use of coupling capacitors in CE amplifier?	CO2	
	a. Block DC		
	b. Pass AC		
	c. Reduce distortion		
	d. Pass AC and Block DC		
3	What is the correct phase shift in CE amplifier?	CO1	1
	a. 45 Degrees		
	b. 60 Degrees		
	c. 120 Degrees		
	d. 180 Degrees		
4	What is the main advantage of FET which makes it more useful in industrial applications?	CO2	1
	a. Voltage controlled operation		
	b. Less cost		
	c. Small size		
	d. Semiconductor device		
	N channel Depletion MOSFET is known as	CO1	
	a. Normally ON then OFF		
	b. Normally OFF then ON		
	c. Normally ON		
	d. Normally OFF		
6	P channel Enhancement MOSFET is known as	CO2	1
	a. Normally ON then OFF		
	b. Normally OFF then ON		
	c. Normally ON		
	d. Normally OFF		
7	Power amplifier directly amplifies _____	CO1	1

	a. Voltage of signal	b. Current of the signal	c. Power of the signal	d. All of the mentioned		
8	Where does the Q point lie for class B amplifier?				CO2	1
	a. Active	b. Cut off	c. Saturation	d. Between saturation and active		
9	Bandwidth due to negative feedback amplifier				CO2	1
	a. Increases	b. Decreases	c. Remains same	d. Partially changes		
10	Gain due to negative feedback amplifier				CO1	1
	a. Increases	b. Decreases	c. Remains same	d. Partially changes		
11	According to Barkhausen criteria of oscillation total phase shift should be-----Degrees				CO1	1
	a. 0 or 360	b. 180	c. 270	d. 90		
12	Which pin number is the input for 7905 IC				CO2	1
	a. 1	b. 2	c. 3	d. 4		
Q.2	Solve the following.					12
A)	Explain need of biasing and voltage divider biasing method				CO1	6
B)	Explain in detail effects of bypass and coupling capacitor				CO2	6
Q.3	Solve the following.					12
A)	Explain JFET and its characteristics, explain the FET as VVR.				CO2	6
B)	Explain in detail D MOSFET and E MOSFET				CO1	6
Q.4	Solve Any Two of the following.					12
A)	Explain class A power amplifier and derive expression for efficiency				CO2	6
B)	Explain class B push pull power amplifier and derive expression for efficiency				CO1	6
C)	Explain crossover distortion and need of heat sink, its design				CO1	6
Q.5	Solve Any Two of the following.					12
A)	Derive expression for voltage gain with feedback for both positive as well as negative feedback amplifier				CO3	6
B)	Explain with block diagram types of feedback amplifier				CO1	6

C)	Explain effect of negative feedback amplifier on input, output impedance, voltage gain, current gain, bandwidth, noise and distortion	CO2	6
Q. 6	Solve Any Two of the following.		12
A)	Explain Barkhausen criteria of oscillation and RC phase shift oscillator	CO1	6
B)	Explain and derive expression for oscillation frequency of Wien bridge oscillator	CO2	6
C)	Explain in detail transistorized series voltage regulator	CO2	6
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